

```
import pandas as pd
import numpy as np
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import LabelEncoder
from sklearn.ensemble import RandomForestClassifier
from sklearn.metrics import accuracy_score, classification_report
```

```
df = pd.read_csv("Dataset .csv")
df.head()
```

	Restaurant ID	Restaurant Name	Country Code	City	Address	Locality	Locality Verbose	Longitude	Latitude	Cuisines	...	Curre
0	6317637	Le Petit Souffle	162	Makati City	Third Floor, Century City Mall, Kalayaan Avenu...	Century City Mall, Poblacion, Makati City	Century City Mall, Poblacion, Makati City, Mak...	121.027535	14.565443	French, Japanese, Desserts	...	Botsw Pul
1	6304287	Izakaya Kikufuji	162	Makati City	Little Tokyo, 2277 Chino Roces Avenue, Legaspi...	Little Tokyo, Legaspi Village, Makati City	Little Tokyo, Legaspi Village, Makati City, Ma...	121.014101	14.553708	Japanese	...	Botsw Pul
2	6300002	Heat - Edsa Shangri-La	162	Mandaluyong City	Edsa Shangri-La, 1 Garden Way, Ortigas, Mandal...	Edsa Shangri-La, Ortigas, Mandaluyong City	Edsa Shangri-La, Ortigas, Mandaluyong City, Ma...	121.056831	14.581404	Seafood, Asian, Filipino, Indian	...	Botsw Pul
3	6318506	Ooma	162	Mandaluyong City	Third Floor, Mega Fashion Hall, SM Megamall, O...	SM Megamall, Ortigas, Mandaluyong City	SM Megamall, Ortigas, Mandaluyong City, Mandal...	121.056475	14.585318	Japanese, Sushi	...	Botsw Pul
4	6314302	Sambo Kojin	162	Mandaluyong City	Third Floor, Mega Atrium, SM Megamall, Ortigas...	SM Megamall, Ortigas, Mandaluyong City	SM Megamall, Ortigas, Mandaluyong City, Mandal...	121.057508	14.584450	Japanese, Korean	...	Botsw Pul
5 rows × 21 columns												

```
df = df[['City', 'Price range', 'Aggregate rating', 'Votes',
        'Has Online delivery', 'Has Table booking', 'Cuisines']]
```

```
df['Cuisines'] = df['Cuisines'].fillna('Unknown')
df['City'] = df['City'].fillna('Unknown')
```

```
df['Cuisines'] = df['Cuisines'].apply(lambda x: x.split(',')[0])
```

```
top_cuisines = df['Cuisines'].value_counts().nlargest(10).index
df = df[df['Cuisines'].isin(top_cuisines)]
```

```
df['Has Online delivery'] = df['Has Online delivery'].map({'Yes': 1, 'No': 0})
df['Has Table booking'] = df['Has Table booking'].map({'Yes': 1, 'No': 0})
```

```
df = pd.get_dummies(df, columns=['City'])
```

```
le = LabelEncoder()
df['Cuisines'] = le.fit_transform(df['Cuisines'])
```

```
X = df.drop('Cuisines', axis=1)
y = df['Cuisines']

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42)
```

```
model = RandomForestClassifier(random_state=42)
model.fit(X_train, y_train)
```

▼ RandomForestClassifier ⓘ ?

```
RandomForestClassifier(random_state=42)
```

```
y_pred = model.predict(X_test)
```

```
accuracy = accuracy_score(y_test, y_pred)
print("Accuracy:", accuracy)

print("\nClassification Report:\n")
print(classification_report(y_test, y_pred))
```

Accuracy: 0.368496079828938

Classification Report:

	precision	recall	f1-score	support
0	0.51	0.46	0.48	48
1	0.18	0.12	0.14	115
2	0.23	0.25	0.24	127
3	0.15	0.07	0.10	165
4	0.25	0.16	0.20	50
5	0.10	0.07	0.08	137
6	0.07	0.04	0.05	56
7	0.48	0.69	0.57	598
8	0.07	0.04	0.05	57
9	0.12	0.04	0.06	50
accuracy			0.37	1403
macro avg	0.22	0.19	0.20	1403
weighted avg	0.30	0.37	0.32	1403